

A Naval Integrated Power System with a Battery Energy Storage System

Fuel efficiency, reliability,
and quality of power.



MECHANICAL MACHINERY HAS DOMINATED THE FIELD of ship propulsion systems in the last century. Since fuel accounts for a large portion of the ship life-cycle cost, the current trend of moving toward fuel economy and environmental regulations in marine engineering has led to the efficient use of onboard energy. Incorporating power electronics into ship propulsion systems has been a very common method of improving fuel efficiency since approximately 1980. One high-profile example is the 1968-built Queen Elizabeth 2, whose steam turbines were replaced with a diesel-electric propulsion plant in 1986. Some commercial ships are now being built with electric drives, such as passenger ships, car ferries, shuttle tankers, cable-laying ships, icebreakers, and floating offshore platforms. Navies around the world have been actively exploring integrated power systems (IPSs) for use on future surface combatant ships since the late 20th century. Although the ship IPS has many distinct advantages, such as increased survivability and maneuverability, reliability inhibits further fuel economy improvement. The onboard battery energy storage system (BESS) was recently

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suggested to increase fuel economy and ensure reliability at the same time. Furthermore, an active front-end (AFE) converter can provide improved grid quality at transient states such as fault and future mission load in addition to steady state as with normal sea sailing.

The Advantages of an IPS for Naval Surface Ships

Before the ship IPS was introduced, most naval surface ships employed a conventional power system with a typical structure that is shown in Figure 1(a). This conventional power system has two subsystems. One subsystem is for ship propulsion with prime movers, reduction gears, long shafts, and propellers. The propulsion power supplied from the prime movers flows through mechanically connected reduction gears and shafts and finally to the propellers. The usual prime mover sets consist of diesel engines (DEs) and/or gas turbines (GTs) whose configurations vary according

to the role of the ship and other issues such as cost and space. A series of reduction gears appropriately slow down the propeller shaft from a relatively high engine shaft while transferring propulsion power. The other subsystem is for ship service loads whose power supply comes from the engine-generator sets. The engines of each subsystem can be used only for their respective purpose since each is physically separated from the other. This typical separated configuration has been used to secure maneuverability, which locks almost 90% of the available power into the propulsion system. However, this exclusive use of the power supply only to the propulsion system can be a disadvantage, as the power is not available for other mission loads. The electric mission loads becomes essential like the maneuverability, but its power demand cannot be met with conventional ship service load supplies.

As an alternative to the conventional power system, the ship IPS was introduced. It has multiple integrated



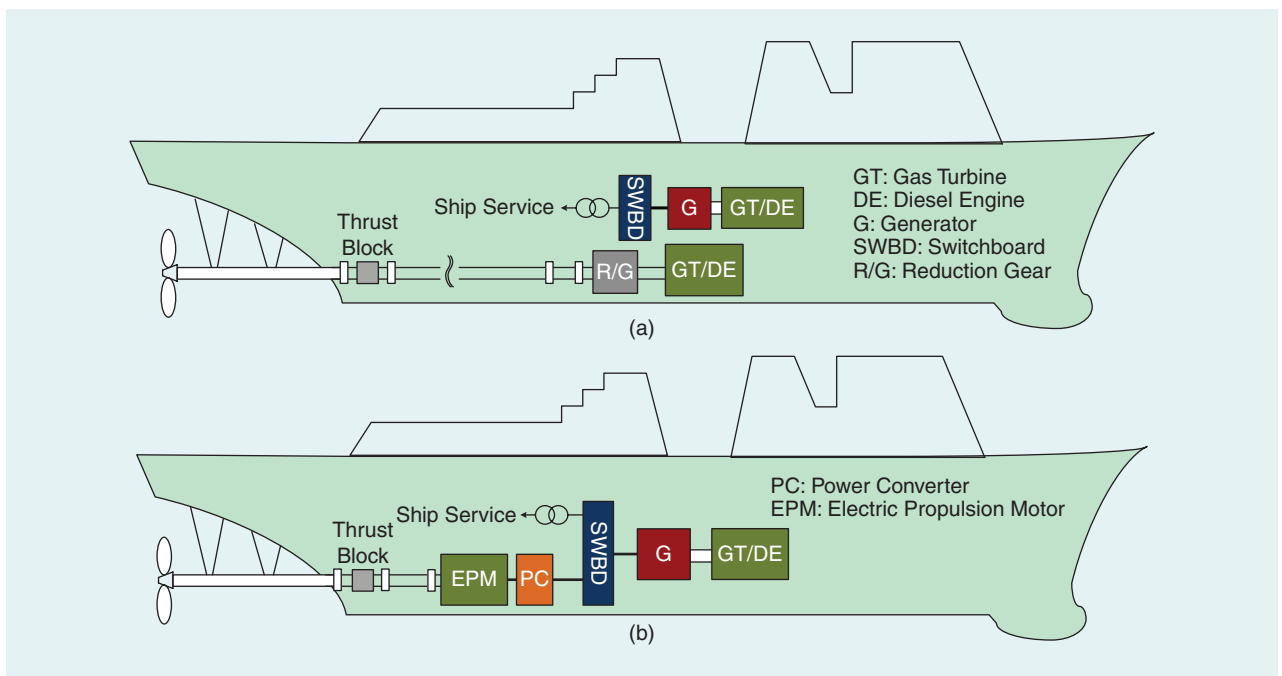


Figure 1. Schematic diagrams of naval ships with (a) a conventional power system and (b) an IPS.

engine-generator sets that share power electrically with ship service and weapon system loads and employ an electrical motor with a power converter for the propulsion, as shown in Figure 1(b). The greatest advantage of shipboard IPS comes from power sharing on the common electrical grid, which improves power flexibility and availability. The advantages of IPSs for naval surface ships are summarized as follows:

- 1) The IPS can provide the same power with fewer prime movers, which enhances the fuel efficiency conspicuously over low- and medium-speed ranges compared to the conventional mechanical power system. A typical warship equipped with GTs can have high power density but would have very poor fuel efficiency at light load operating conditions. The IPS can save fuel with an optimum number of generators, depending on the specific circumstances of the ship. The hybrid electric drive is currently receiving a considerable amount of attention for achieving operational efficiency in warships.
- 2) The IPS can increase the naval ship survivability because of its reconfigurable power grid and the enhanced reliability from the redundant structure. System redundancy is an essential requirement for combatant ships in case of any faults and recovery. The IPS could easily achieve system redundancy by separation, cross-wiring, or fail-safe operation of essential equipment.
- 3) The surface ships with IPSs can extend the electric power supply for future weapon systems such as high-energy lasers or electromagnetic guns. The power demands of these advanced weapon systems can be up to tens of megawatts, possibly as a series of

pulses with very short time intervals. A shipboard electrical architecture that supports these weapons with a separated power source would be unaffordable because of the additional weight and volume of the power source.

- 4) The ship IPS electric propulsion system can enhance the maneuverability based on much faster electrical motor dynamics driven by a power converter with a fixed pitch propeller rather than the conventional long shaft with a controllable pitch propeller. The electrical motor could force reverse thrust and allow a ship to crash stop at a shorter distance.
- 5) The IPS can provide a more comfortable residential environment to the crew, and it needs less maintenance because of the inherent property of the electrical system when compared to the mechanical system. Furthermore, the united control and monitoring system can be achieved only with a local and remote place, possibly with a remarkably reduced number of operating crew members.

A Conflict Between the Power Reliability and the Fuel Efficiency

Reliability, which can be expressed as a fail-safe operation, is the most desired property for a naval surface ship. Even if an IPS component fails, the IPS should remain unharmed and in operation. Securing reliability involves system redundancy, as shown in Figure 2, where a typical naval ship IPS model with dual redundant main buses and ship service buses is shown. The generation system consists of multiple generators that operate in parallel for a fail-safe operation. The main switchboards, or point of

common couplings (PCCs) of the grid, are physically separated to each other as well as their own tie circuit breakers to enhance the chances of survival. There are two propulsion systems for redundancy and survivability. Furthermore, half of the propulsion power is provided by its own switchboard and the other half by the opposite bus for cross-feeding, which is not described in Figure 2. Therefore, even if one switchboard fails for any reason, continuous power can be provided to the propulsion motors.

However, this type of reliability is a hindrance for fuel economy. The engine generators run for reliability seldom result in efficient operation points, which usually lie above the medium load. This tradeoff between reliability and fuel economy becomes more complicated as the naval surface ship operates in various speeds and the electric loads vary independently. The minimum generator operation (MGO) with N-1 safety (contingency) can be considered as a simple solution and, further, as a standard operation rule to this tradeoff. This rule is based on the idea that the less the generators are run, the more

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load is on each generator. This, in turn, makes the operation more efficient. However, the predetermined system components and operation status mean that there is not much room for improvement.

Adopting an onboard BESS provides an opportunity for improvement. The BESS can be seen as an additional degree of freedom in the system design. Hence, the fuel efficiency can be increased while guaranteeing reliability. When one online generator suddenly stops for any reason, the BESS can cover the active power shortage for a while to keep the power grid continuously maintained like an extra generator

until another generator on standby is synchronized according to N-1 safety. Therefore, the BESS can change the operation strategy of the engine generator to become more efficient without sacrificing reliability.

Design Procedure of the Onboard BESS Power Capacity

The power capacity and energy of the onboard BESS should be determined according to MGO and N-1 safety.

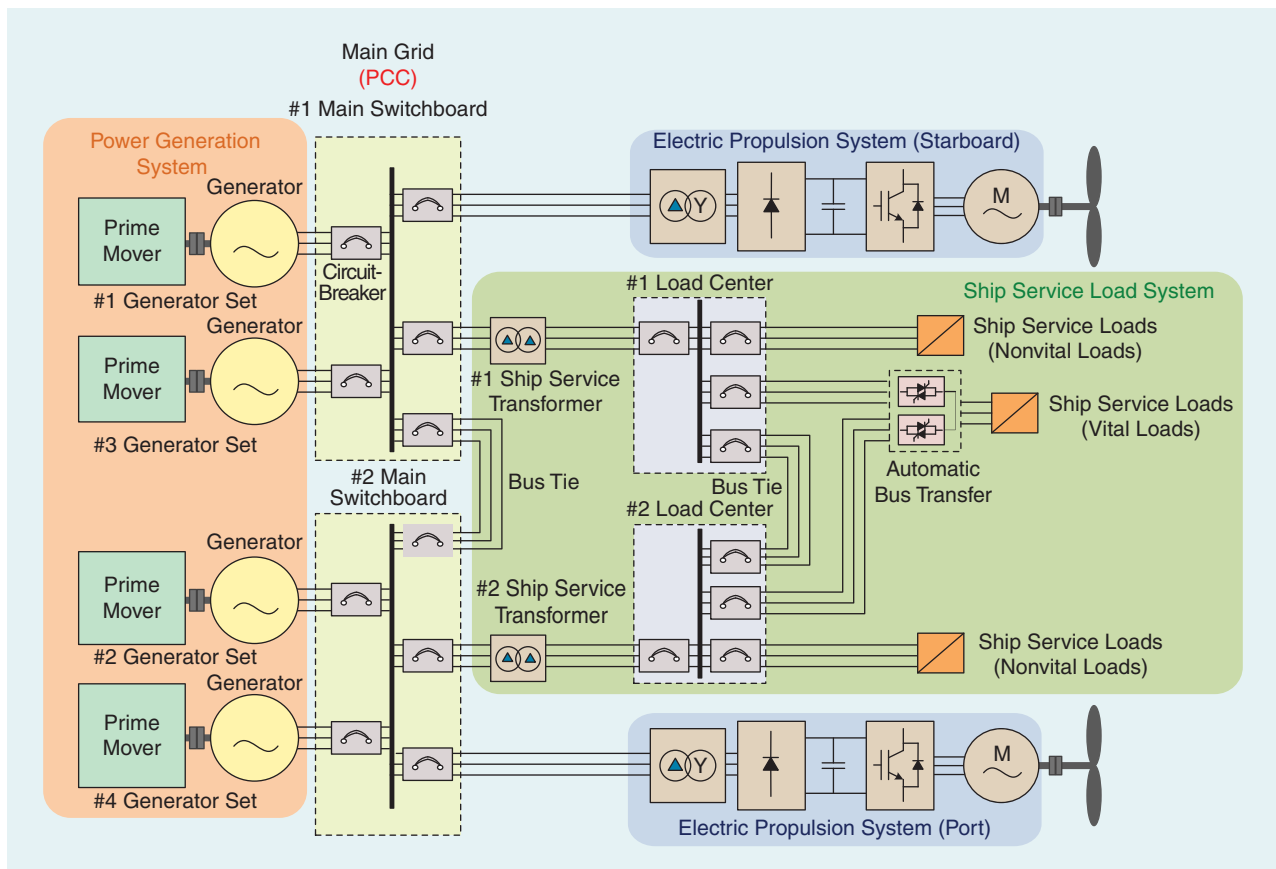


Figure 2. A typical IPS of a naval ship model (small and medium class).

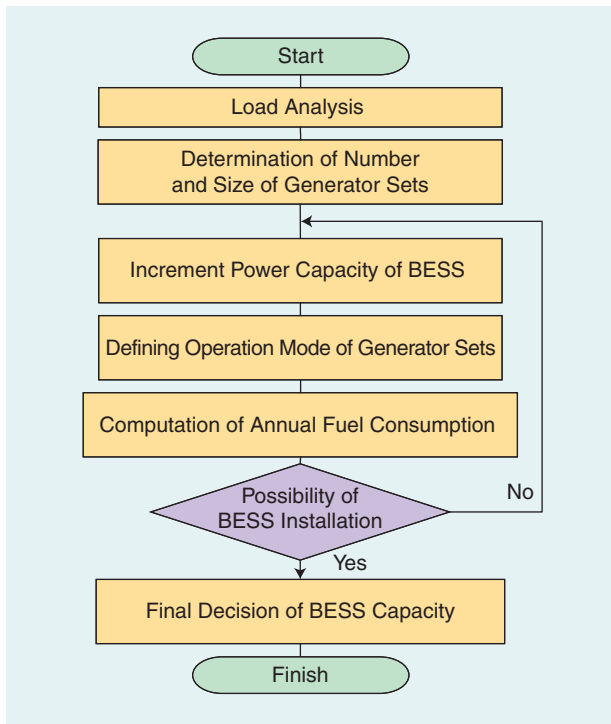


Figure 3. The flow chart to design the BESS capacity.

The first procedure related to N-1 safety determines the allowed engine-generator sets, while the second procedure related to MGO calculates the fuel savings and BESS cost. Finally, the BESS capacity is determined after the iterations of these two procedures, as shown in a flow chart in Figure 3.

The N-1 safety completion can be distinguished by the load factor for safe operation, LF_{safe} , according to the online generators as

$$LF_{safe} = \frac{P_{G,total} - P_{G,trip}}{P_{G,total}} \times 100, \quad (1)$$

Table 1. An Example of Engine-Generator Combinations for Safe Operation (x Corresponds to 1 or 2).

Combination of Engine-Generators	Total Power ($P_{G,total}$, MW)	LF_{safe} (%)	Sustainable Load (MW)
#1 (SDGx)	1	(N.A.) 0	0
#2 (SDG1, 2)	2	50	1
#3 (LDGx)	2.5	(N.A.) 0	0
#4 (SDGx+ LDGx)	3.5	28.6	1
#5 (SDG1, 2 +LDGx)	4.5	44.4	2
#6 (LDG1, 2)	5	50	2.5
#7 (SDGx+ LDG1, 2)	6	58.3	3.5
#8 (SDG1, 2 + LDG1, 2)	7	64.3	4.5

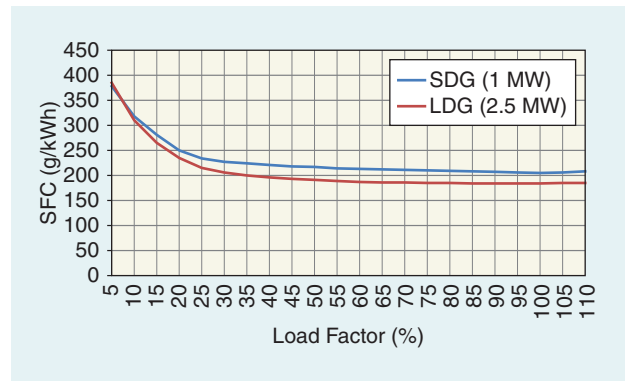


Figure 4. The SFC curves of engine generators.

where $P_{G,total}$ is the total generation power of the online generators and $P_{G,trip}$ is the power loss caused by the largest online generator failure. LF_{safe} should be a positive number for a fail-safe operation, and the larger LF_{safe} refers to higher fuel efficiency. The output power of each generator P_{Gi} is restricted as in (2) under the assumption of a same load ratio among engine generators regardless of their capacity

$$P_{Gi} \leq LF_{safe} \times P_{Gi,max}. \quad (2)$$

The operation concept can be explained with an example ship constructed by the Republic of Korea Navy. This ship has four DE-generators (DGs): two 1-MW small DGs (SDG1, SDG2) and two 2.5-MW large DGs (LDG1, LDG2). Every possible engine generator combination is listed in Table 1. Sustainable load stands for the maximum sustainable load after the loss of the largest online generator. For the fail-safe operation, a load connected to the grid cannot exceed the corresponding sustainable load value of each combination. For example, #5 can be in operation only when the load is less than 2 MW. Among the eight combinations, #1 and #2 cannot be allowed because of zero LF_{safe} . Also, #4 is not a good option because it has the same sustainable load as #2 but a smaller load factor. The allowed combinations are shaded, and the operation region of each combination lies within its sustainable load values while the available maximum load factor for the fuel economy is maintained. According to this principle, #5 can operate with the load between 1 and 2 MW. Because a load of more than 4.5 MW cannot be sustained against engine-generator failure, appropriate and immediate load shedding is necessary to keep the IPS in operation.

The presence of an additional power source such as the BESS

changes the total power LF_{safe} and sustainable load in Table 1. As the BESS capacitor increases, the allowed types of combinations change and their corresponding LF_{safe} changes. The engine generator has a better fuel efficiency at the higher load ratio whose typical specific fuel consumption (SFC) curve is shown in Figure 4, and this is the same as the higher LF_{safe} . Therefore, the saved fuel cost can justify the additional BESS cost, weight, and volume in certain cases.

The estimated annual fuel savings based on the annual operating profile of the naval ship depend on the BESS capacity, as shown in Figure 5, as well as the BESS payback period and weight, as shown in Figure 6. In this estimation, the cost and weight of onboard BESSs are assumed to be US\$1,800/kWh and 60 kg/kWh. This is approximately two and three times larger, respectively, when compared with those currently available in commercial applications because of military specifications. The maximum continuous discharging current of the battery module is assumed to be 4C-rate. As the BESS capacity increases, the possible fuel savings increase as well, but the tendency is saturated above a certain point. Based on the assumptions, 1,000 kW/250 kWh of the BESS might be selected to have a reasonable payback time with an affordable BESS weight. These resulting figures and selection may change as the specific configuration of the ship IPS and operating profile change. The saved fuel cost would be much larger for the IPS with the GT because of its poor fuel efficiency at lower loads.

Circuit Configuration and Control Strategy of Onboard BESS: Direct Connect to the DC-Link of the Electric Propulsion System

There are a variety of circuit configurations and control strategies of the BESS as used in widespread applications. The circuit configurations have prime design factors such as 1) node location where the BESS connected with, 2) (its) hardware structure, and 3) the corresponding control method. They are subject to cost, space, efficiency, reliability, and others. Many of the applications, which employ the BESS to support grids similarly to our purpose, interfaced the BESS with the PCC through its own power converter or placed the BESS on the dc-link of the propulsion system through the dc/dc converter. However, the BESS can be directly connected to the dc-link of the propulsion system, as shown in

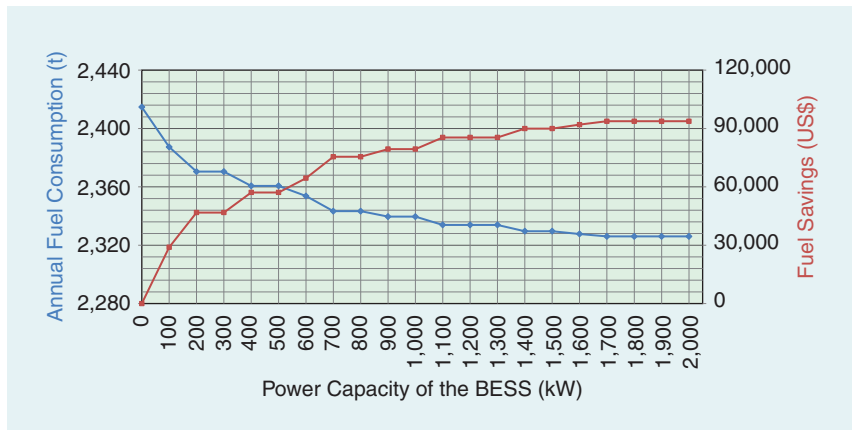


Figure 5. The annual fuel savings according to the capacity of the BESS.

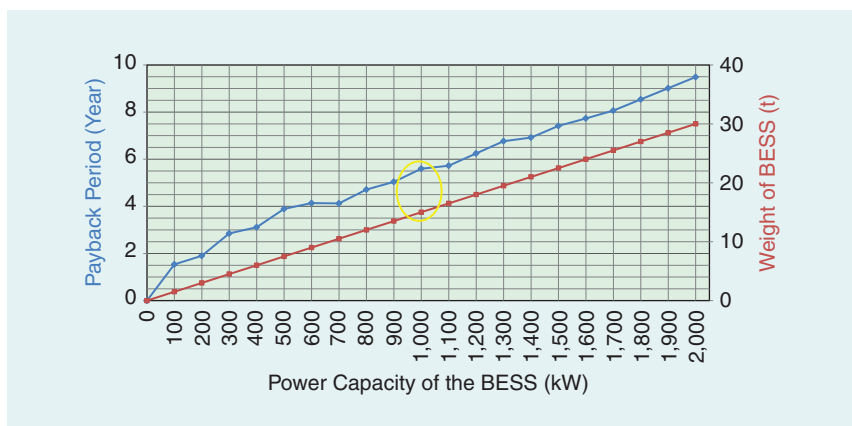


Figure 6. The payback period and system weight for BESS installation according to the capacity of the BESS.

Figure 7, eliminating the cost, weight, volume, and loss of the dc/dc converter. At the same time, the 12 pulse diode rectifiers with a bulky transformer in Figure 3 can be replaced with an AFE converter, as shown in Figure 7. This can improve the harmonic characteristics and provide the bidirectional power flow capability. The cost of this configuration is that the BESS voltage, which varies according to the state of charge (SOC), should be accommodated by the propulsion inverter and the AFE converter. This would result in the oversizing of the inverter and AFE converter.

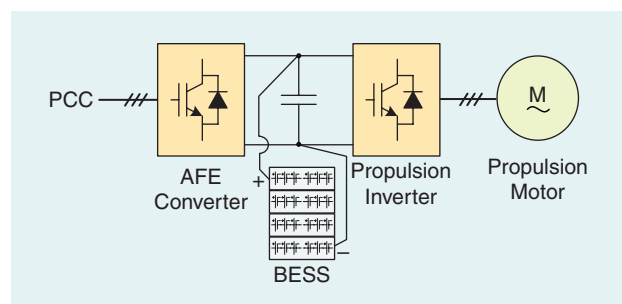


Figure 7. A circuit configuration diagram of the BESS direct connection to the dc-link of the propulsion system.

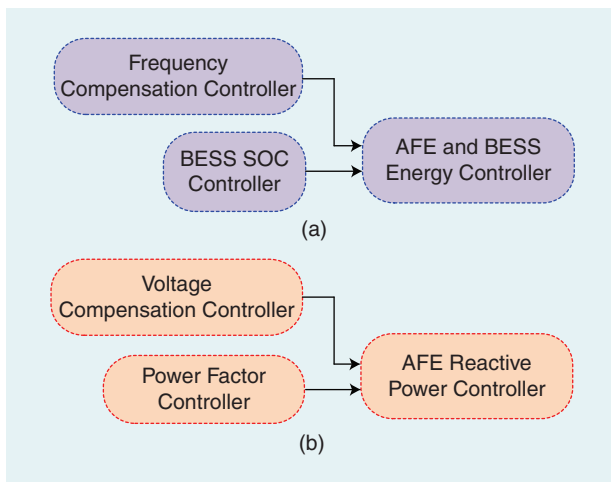


Figure 8. The AFE rectifier control strategies of (a) active and (b) reactive power.

Because the majority of the power is consumed by the propulsion motor, the configuration in Figure 7 may provide a better BESS operating efficiency.

The power control of a directly connected BESS can be achieved because the other components connected to the BESS firmly determine their respective power flow. The AFE converter determines the power flow between the PCC and dc-link. The propulsion inverter determines the power flow between the propulsion motor and the dc-link. Hence, the mismatched power inevitably flows to the BESS. The propulsion inverter maneuvers the ship, so it consumes or regenerates active power only to secure the maneuverability regardless of the other ship IPS components' status. Therefore, indirect control of the BESS is accomplished by the AFE converter. The active and reactive power control strategies of the AFE converter are shown in Figure 8. The AFE converter transfers active power from the PCC to the propulsion inverter most of the time, keeping the BESS SOC in a certain range. Since there is normally no need of reactive power flow, it keeps the power factor as unity. In abnormal situations such as generator trip or faults, the PCC frequency or voltage can deviate from its recommended ranges. The frequency and voltage compensators counteract this abnormal PCC behavior, suppressing transients and keeping both the voltage and frequency within tolerable ranges by controlling the active and reactive power flows. There would be dead bands in the compensators not to disturb the governor (GVR) and automatic voltage regulators (AVRs) of generators in normal situations.

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Experimental Test Setup of a Small-Scaled, Real-Time Power-Hardware-in-the-Loop Simulation

As a ship IPS construction requires a considerable amount of time, cost, space, etc., a preceding IPS simulation is highly recommended. However, the ship IPS complexity makes it difficult to model and simulate properly. The typical IPS structure, as shown in Figure 2, contains engine generators with associated controllers, switchboards, ship service loads, propulsion loads with BESSs and pulsewidth modulation (PWM) converters, future mission loads, and system faults. The typical simulation model in Figure 2 with the aforementioned BESS configuration, as depicted in Figure 7, is shown in Figure 9. The model has simplified the ship service load to constant resistor and inductor (RL) loads because the RL load variations are slow enough to be handled with classical engine generators and associated controllers: governors and AVRs. The propulsion system is modeled as the ideal current source, which neglects the switching and control dynamics of converters to reduce the simulation time. The BESS is modeled as simple resistor, inductor, and capacitor and dc voltage source circuits whose physical limitations are ignored. The future mission load, which is in the form of the pulse load, is modeled as the current sink interfaced with Y-Y, Δ transformer, and diode rectifiers. This simulation model can be useful for system configura-

tion design levels such as when deciding the component capacity, placement and wiring, and steady-state analysis.

As the simplified model has bounds related to the detail characteristics of an actual system, more detailed models of each component should be considered. The detailed propulsion system model includes an AFE converter, a BESS of hundreds of submodules, a PWM inverter, a propulsion motor, and ship hydrodynamics. The propulsion system includes ship hydrodynamics and Robinson curve, as shown in Figure 10. The Ship Hydrodynamics Model and Propulsion Motor Plant are mechanical models and propulsion motors related to their shape, structure, etc. The Robinson curve

is a two-dimensional input function according to the propeller shape, which describes the relationship among the propeller (fixed with propulsion motor shaft) speed, ship speed, and ship thrust. The ship speed/thrust controller and motor speed controller belong to the control strategy of the propulsion inverter, which can vary as the control strategy changes. The ship speed follows its commanded value with a typical time constant of several seconds to tens of seconds. The PWM inverter and the AFE converter usually have a few kilohertz switching frequency. Thus, a proper time step of simulation should be less than 10 μ s. This short time step and large time constant of the ship hydrodynamic system causes stiff system problems since simulations should last more than tens of seconds for a few

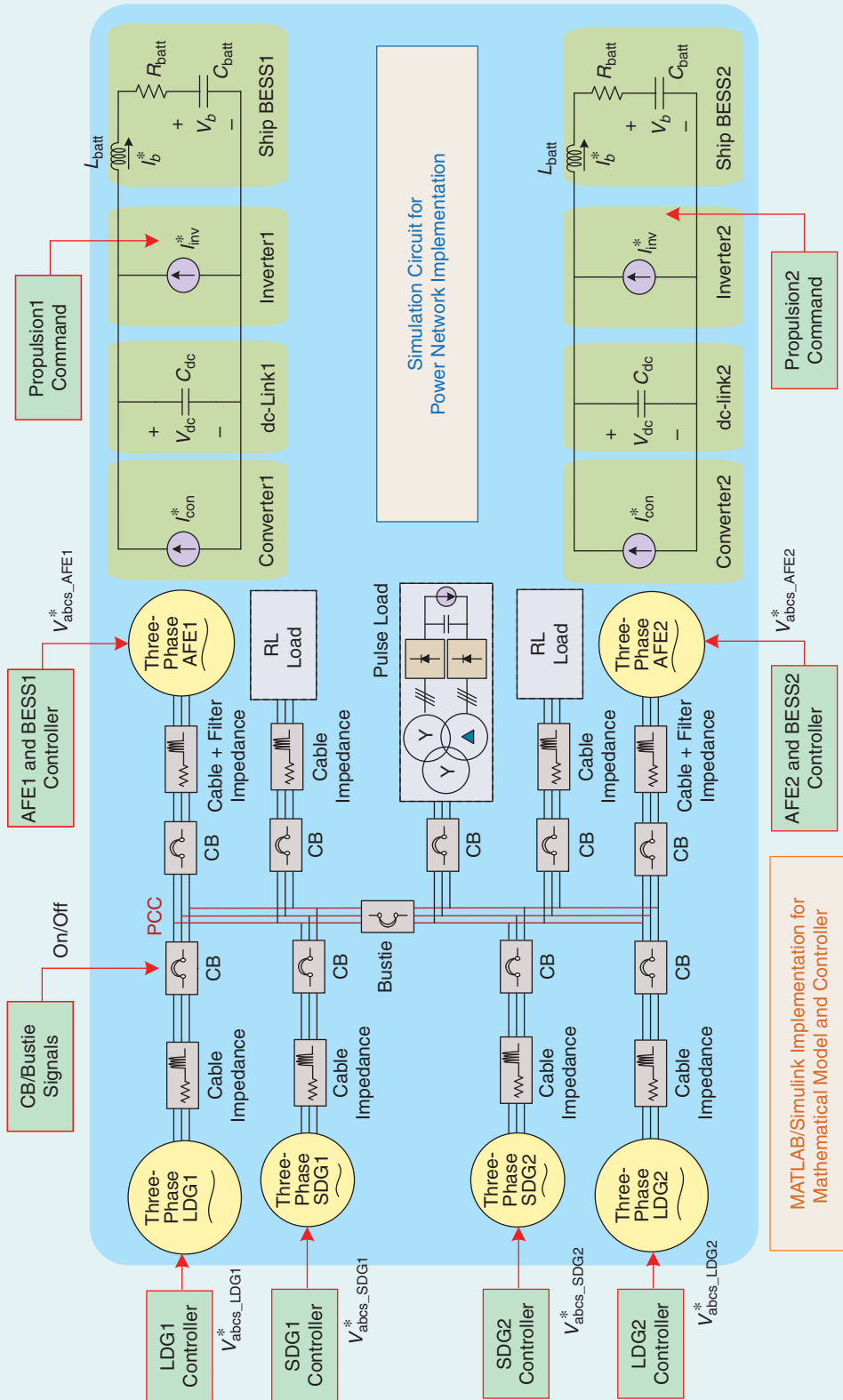


Figure 9. The simulation model of a naval IPS with an onboard BESS. CB: circuit breaker.

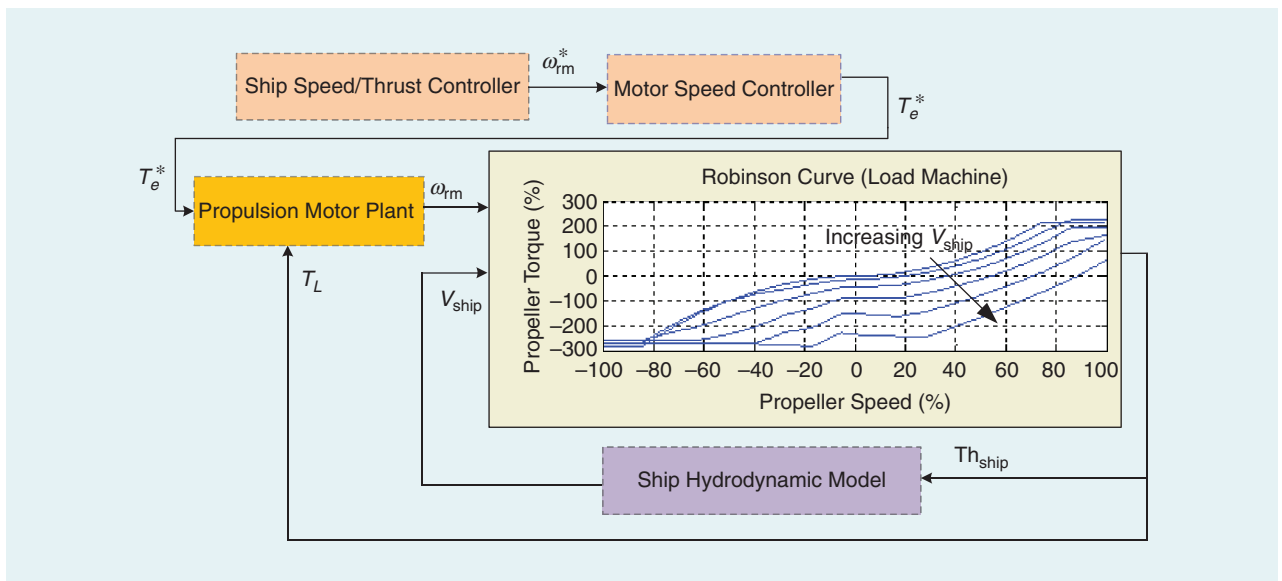


Figure 10. A propulsion system diagram for the PHILS.

microsecond time steps. Hence, the average voltage model of the PWM converter is commonly used. This model assumes a constant output voltage during the switching time so that the time step can be the same as the hundreds of microsecond sampling (switching) period. This simulation model can be useful for relatively slow dynamic system levels such as ship maneuverability, excluding crash stop.

The aforementioned simulation model verification does not involve practical issues such as switching harmonics and ripples and the nonlinear property of the BESS. Modeling these stiff or nonlinear properties is not easy. Hence, the power-hardware-in-the-loop simulation (PHILS), or down-scaled PHILS, is performed. The PHILS setup, shown in Figure 11, consists of software models and down-scaled hardware implementations of a ship IPS. The components that need a small step time or have a higher nonlinear property should be implemented in the hardware, such as the AFE converter, the BESS, the PWM converter, and the propulsion motor. The components that lack accessibility or safety should be modeled as software. This includes engine generators and their associated controllers, pulse loads, propellers, Robinson curve, and ship hydrodynamics.

There are several considerations for the connection of the software and hardware. One consideration is the synchronization between the simulation model and the actual hardware component. The simulation should be run in real time just as with the hardware. As a low-cost solution, a multicore processor can be used to simulate partitioned software models based on a well-known ideal transformer model. In addition, the data transfer between software and hardware must be considered. The one-way transfer from hardware to software can be done easily with sensors and communication, while the returning transfer from software to hardware needs additional high-performance voltage amplifiers to closely imitate the PCC. An example of the amplifier is a high-frequency switching multi-

level inverter with an inductor, capacitor, and inductor (LCL) filter, which can cover the fault situation transient due to its extended voltage regulation bandwidth. Furthermore, the electrical and mechanical dynamics between full- and down-scaled systems must be preserved. The power, torque, speed, inertia, voltage, current, impedances, and other values should remain the same per unit. Some of the parameters can be kept, while others need virtual parameters after proper conversion. This PHILS verification validates most control system levels such as the current and voltage harmonics as well as fast dynamic situations including the crash stop, pulse, and faults.

Ensuring IPS Grid Power Quality with Onboard BESS

The onboard grid power quality is currently an important issue, especially in naval ship IPSs. Ship crews suffer from various power quality problems such as voltage dips, blackouts, abnormal changes in the voltage and frequency, electrostatic discharges, electromagnetic fields, flicker, and harmonics. The AFE convert and BESS of naval ships enhances both the grid voltage and frequency quality simultaneously while suppressing the grid current harmonics. Their operation can be effective at steady load conditions as well as at transient load conditions. This article verifies some ship load conditions through the experimental results of the PHILS. The transient load example involves an active power load that causes sudden and severe grid-frequency changes, such as a generator trip, and an active and reactive power load that causes sudden frequency and voltage changes such as a pulse load.

For example, SDG1 in Figure 9 is tripped during the MGO mode when LDG1, SDG1, and SDG2 are in line with an almost 90% load factor in an outage. The total capacity of all online generators is 4.5 MW, and the total loads demand is 4 MW. Thus, SDG1 trip overloads the system. As shown in Figure 12(a), the PCC frequency drastically drops within a

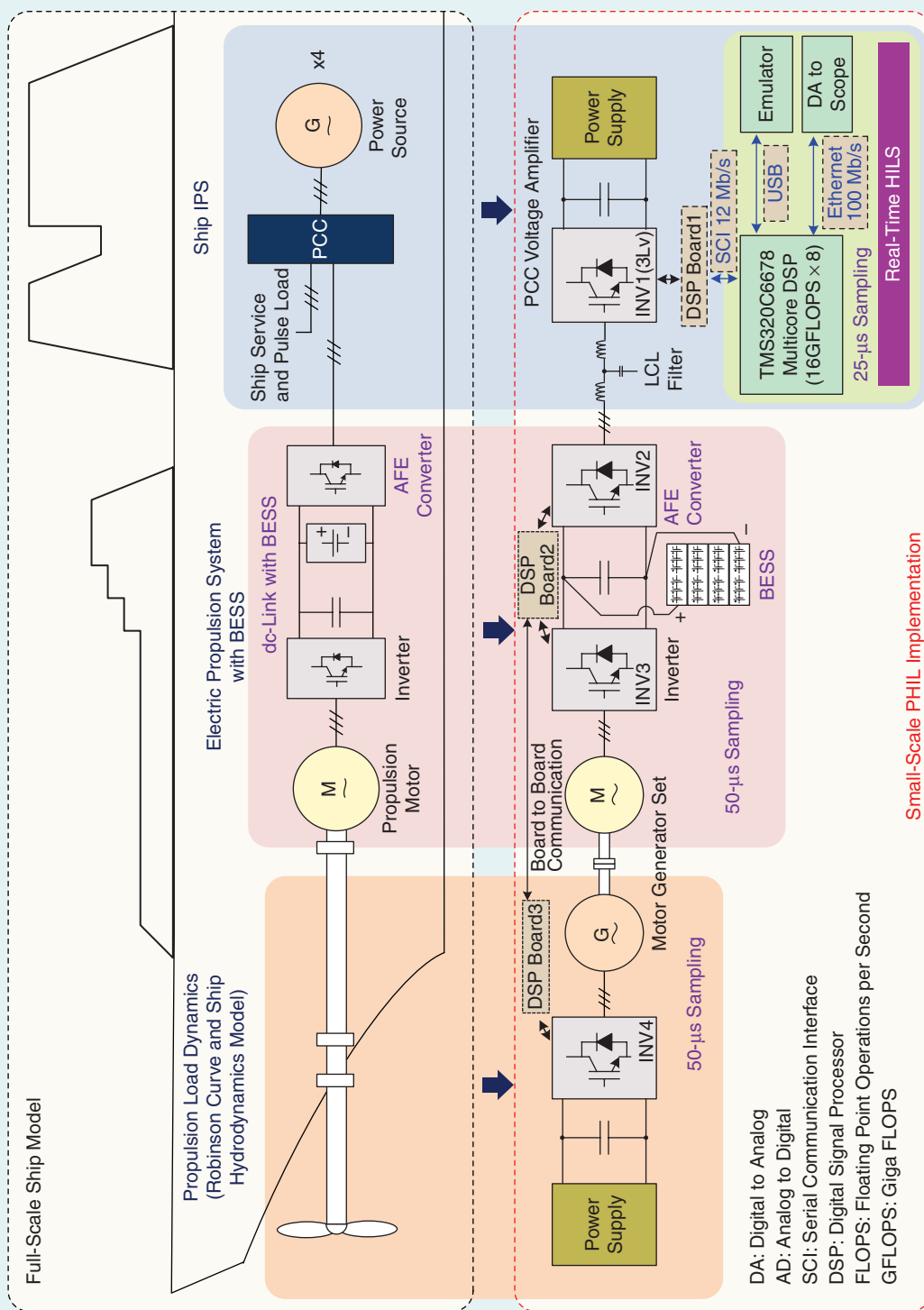


Figure 11. The PHILS setup of a ship IPS with the propulsion load.

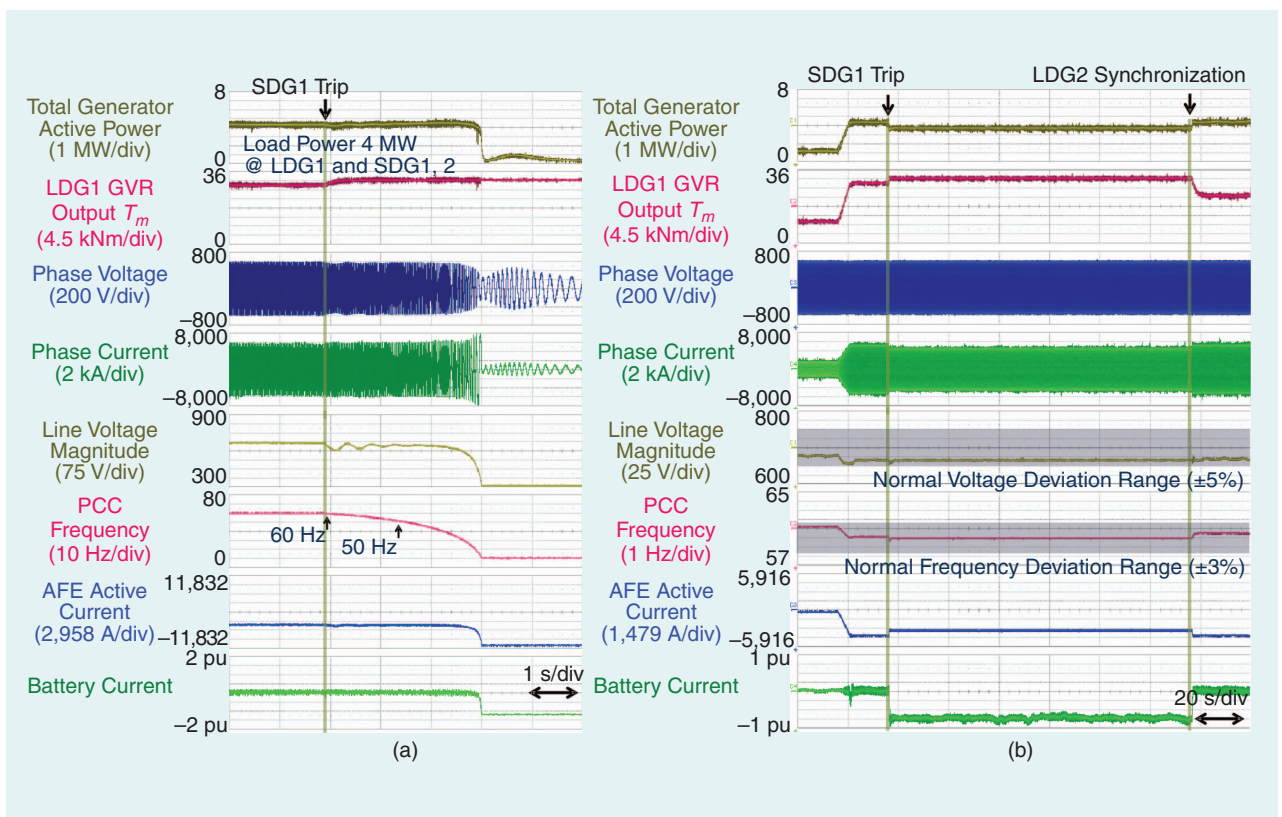


Figure 12. The experimental results of a generator trip (a) without and (b) with BESS.

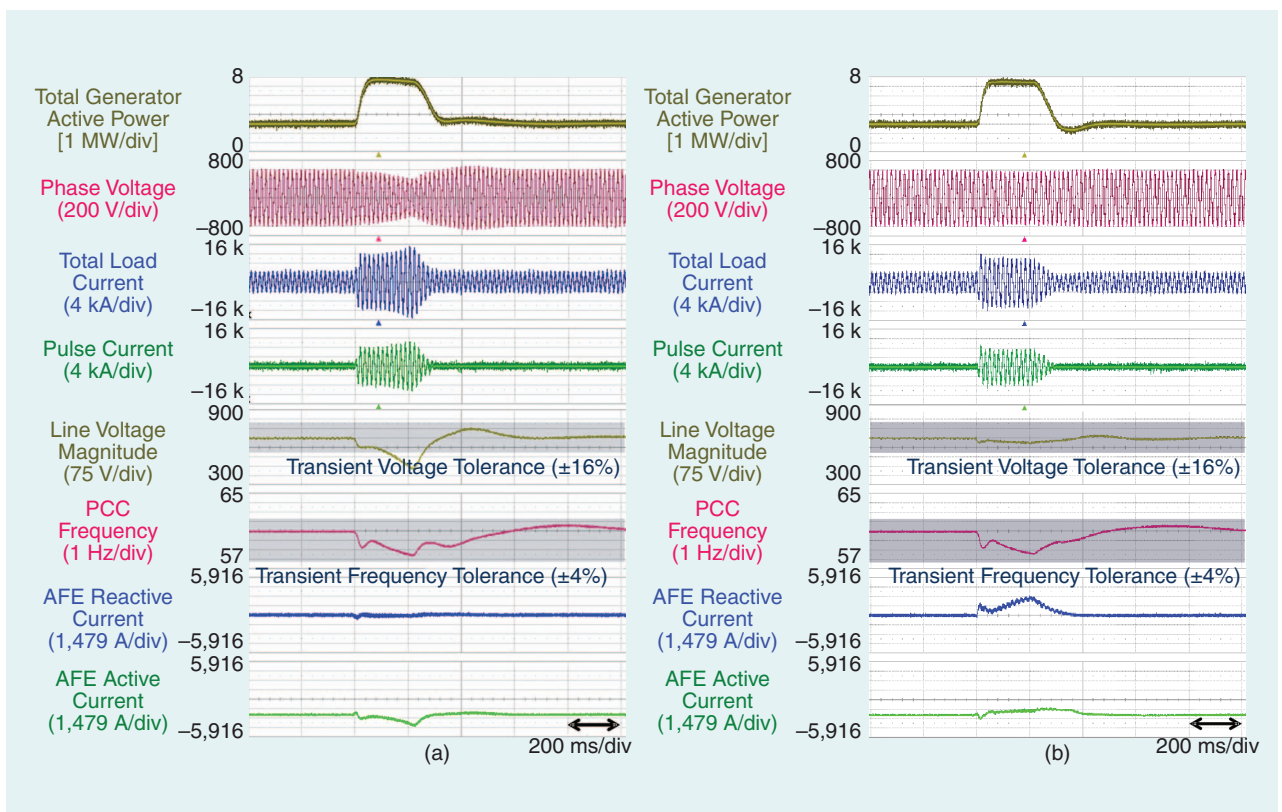


Figure 13. The experimental results of a pulse load (a) without and (b) with compensation control.

few seconds and the system eventually blacks out. Figure 12(b) shows the experimental results of an active power compensation with a 1,000-kW BESS. The BESS current increases, and the PCC frequency is maintained within the normal deviation region specified as $\pm 3\%$ from the nominal value, 60 Hz. The active AFE converter current decreases during the generator trip mode because of the BESS power support. A standby generator, LDG2, is brought to the PCC after 120 s, which is the typical synchronizing time of a cold standby generator. After discharging ends, the BESS SOC and dc-link voltage are restored with the constant charging energy from the engine generators.

For future mission loads, a trapezoidal pulse load was activated, which consumes 5 MW during 200 ms: increasing in 20 ms and decreasing in 80 ms. The ship-service load consumes 1 MW and 250 kVAR, and the propulsion load consumes 2.17 MW. The total active load, including pulse load, demands 8.17 MW. This exceeds the total generation capacity, 7 MW, resulting in an inevitable decrease in the PCC frequency. Figure 13 shows the PHILS results against the pulse load. In Figure 13(a), the pulse load without a compensation strategy is shown, where the PCC line-to-line root-mean-square voltage drops from 690 to 440 V and the PCC frequency drops from 61 to 58.3 Hz, violating the naval grid standard. Figure 13(b) shows the situation with the compensation strategy where the PCC voltage deviation was suppressed to less than 10% and the frequency deviation reduced to 2.5% compared to 60 Hz. Thus, both satisfy the naval grid standard. The active and reactive AFE converter current was controlled to counteract the pulse load. The pulse current shape is stable compared to before due to the stabilized PCC voltage with the proposed algorithm. The transient from the pulse load is recovered in a short period of time.

Conclusion

This article examined the high-capacity onboard BESS design and operation that enhances fuel efficiency, reliability, and quality of power in naval IPSs. Despite all of the advantages of IPSs, fuel consumption optimization remains a challenge for naval ships because of the conflict between system reliability and fuel efficiency. Onboard BESSs can be a practical solution, but it is hard to find onboard BESS introductions to commercial and naval ships. This article proposed a methodology for the design and operation of an onboard BESS as well as a BESS capacity decision procedure based on operating the profile and SFC of engine generators. A recently constructed naval ship serves as an example that prioritizes the electric power supply reliability. The actual BESS capacity depends on the installation conditions (i.e., the weight of the system and the investment costs) as well as the annual fuel savings. The BESS circuit configuration that interfaces it to the dc-link of the AFE converter and propulsion inverter without additional dc/dc converters was suggested. The proposed structure has many excellent features when taking the special characteristics of naval ships into consideration. Moreover, many of the problems that arise from the deviation in the frequency and

voltage of the naval IPS can be actively compensated with the help of the AFE converter and the BESS. The performance against the grid abnormality, such as generator trip and pulse load, was simulated through a real-time, small-scale PHILS.

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For Further Reading

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